Adapting IDS to newly labelled attacks

IDS · Classifying network flows as benign or attack types

Initial deployment

Labelled benign traffic
and known attacks

Newly labelled attack

A small labelled sample
of a newly observed attack

Detector update

Recognize the new attack
and preserve earlier quality

XGBoost Comparator: refit on all labelled examples available so far

TabPFN Fixed pretrained weights; labelled examples supplied as context

Comparison · New-attack detection · Retained quality · Update cost

TabPFN · Weights, context and query

Prediction rules pretrained on synthetic tabular tasks · TabPFN-3, arXiv 2026

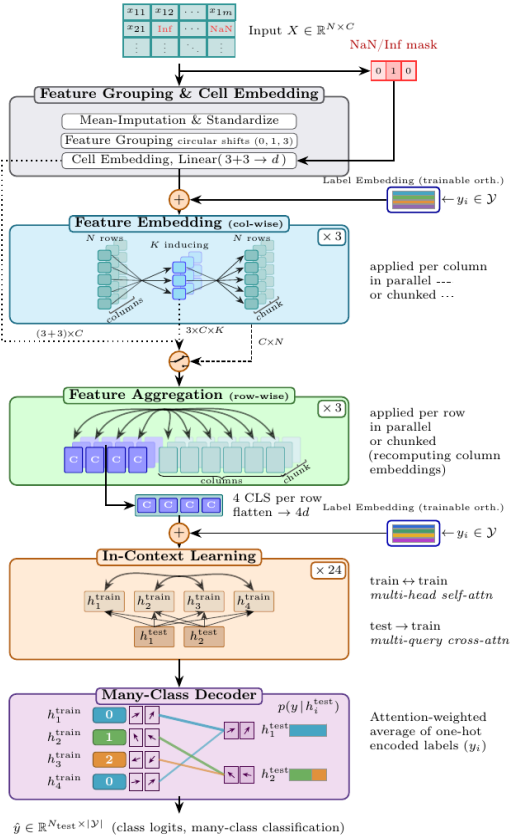


In-context learning Predict the query using its context, with no weight optimization

Fine-tuning Split IDS examples into context/query; update weights by backpropagating the labelled-query loss

Current comparison · Original pretrained weights, with no IDS fine-tuning

v3 architecture · Representing network flows



1 Encode values · Cell embedding

Convert each value, such as packet length, into a vector
Include an indicator for missing values

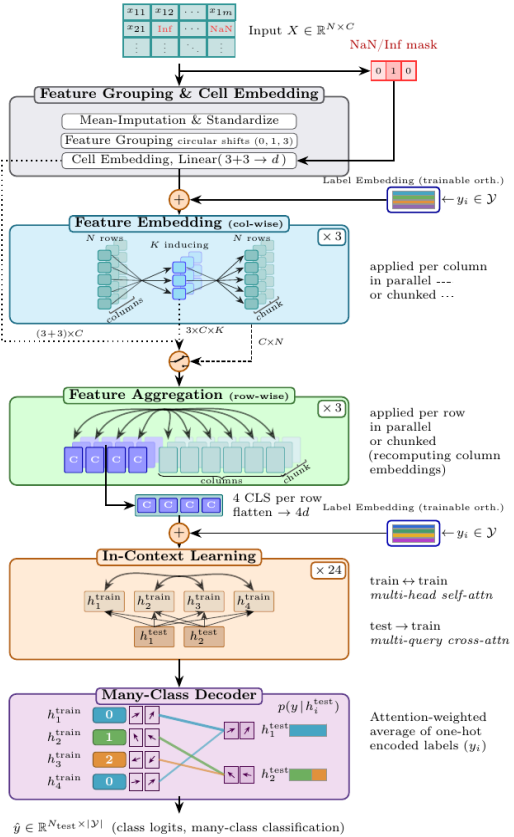
2 Summarize columns · Feature embedding

Summarize each feature across multiple flows
Use the labels supplied with the context

3 Combine features · Feature aggregation

Combine a flow's features into one representation
Prepare it for comparison with other flows

v3 architecture · Context-based prediction



4 Consult examples · In-context learning

Compare the query with labelled context examples
Weight their relevance through attention

5 Predict probabilities · Many-class decoder

Combine context labels with attention weights
Keep examples of both existing and new classes

Context updates also require computation

Preprocessing · Representations · Cache preparation
Measure update time separately from weight training

Evaluation scenario

By providing a series of attack cases in stages, comparing the update costs and detection quality of the two models.

CIC2018

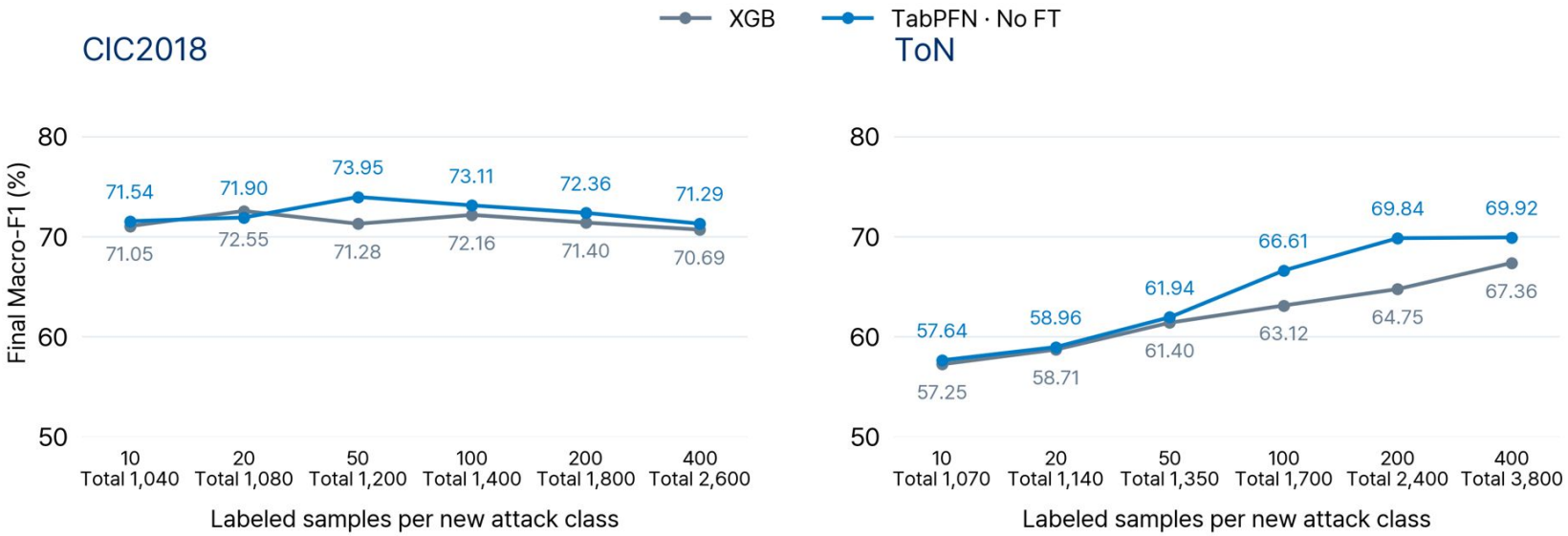
Stage	New classes	Per class	Cumulative
1	Benign Brute force DoS	750 125 125	1,000
2	DDoS	100	1,100
3	Web attacks	100	1,200
4	Infiltration	100	1,300
5	Bot	100	1,400

ToN

Stage	New classes	Per class	Cumulative
1	Benign DoS Scanning	750 125 125	1,000
2	DDoS Injection	100 100	1,200
3	Password	100	1,300
4	XSS	100	1,400
5	Backdoor Ransomware	100 100	1,600
6	MitM	100	1,700

Performance by new attack sample budget

Initial 1,000 fixed · 10/20/50/100/200/400 per new attack class · Final model, full test



400 per new attack · TabPFN minus XGB: CIC2018 +0.60 pp · ToN +2.56 pp

Results

Final model, full test · XGB: cumulative retraining · TabPFN: cumulative context, no FT

